

Muhammad Usman Haider

AI/ML Engineer

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Professional Summary

Experienced **ML Engineer** with **2+ years of experience**, specializing in Generative AI LLMs. Proficient in developing and deploying cutting-edge AI solutions within SaaS and PaaS environments.

Professional Experience

Epazz

Oct 2025 - Present

Product-based company building AI-driven solutions

AI/ML Engineer – Onsite

- Designed and implemented production-grade AI systems, integrating ML/DL models to enhance product intelligence and automate core workflows.
- Developed and optimized LLM-powered solutions, including RAG-based assistants and multimodal AI tools, improving user experience and operational efficiency.
- Scaled deployments with Docker, FastAPI, and AWS, ensuring robust, secure, and high-performing AI solutions in production environments.

STECHAI

Sep 2023 - Sep 2025

A service-based company delivering comprehensive data science, AI, and machine learning solutions.

Associate Data Scientist – Onsite

- Engineered high-performance Django and FastAPI backend services, increasing user traffic by 50%.
- Built and fine-tuned open-source ML and deep learning models (NLP, CV), achieving 15% higher predictive accuracy and automating 65% of repetitive tasks.
- Developed a Retrieval-Augmented Generation (RAG) chatbot using LangChain, Pinecone, and GPT-based models to provide precise legal assistance.
- Deployed the solution using Docker and AWS, ensuring high availability, scalability, and efficient resource utilization in a production environment.

Projects

Retail Knowledge Graph – Led the development of a knowledge graph for retail data by extracting data from MongoDB, designing ontologies and relationships, and storing them in Neo4j. Implemented a question-answering system powered by LangChain agents to enable natural language interaction with the data.

Student Medical Chatbot – Developed a chatbot app for MBBS students to assist with their medical books using Llama Index. Improved accuracy by fine-tuning the model and storing embeddings in OpenSearch on AWS Cloud. Built the backend in Django with proper authentication and session management.

Chatical - I led a project where we built a Chrome extension that automatically captures the content of the current webpage when you open it. The content is then stored in a Pinecone vector database, allowing users to chat and ask questions about the article they're viewing.

TrainMyAI - I develop a user-friendly tool designed to enable anyone to train deep learning models with just a single click. Built with a simple and interactive interface powered by Streamlit, this project aims to simplify the process of model training, evaluation, and inference.

Inventory Analysis – Led the project and trained an XGBoost model on a dataset with over 5 million rows. Developed Django APIs to streamline model training and prediction for both sales forecasting and replenishment optimization. Integrated with AWS for data storage and retrieval.

Football Analysis – Developed a football analysis system that identifies player positions and tracks their movement speed in real time during live matches. We used YOLO for object detection, OpenCV for video processing, and PyTorch for building the underlying models.

AI-Powered Exam Evaluation System - I led a project to develop an offline AI backend that grades scanned student answers, delivers personalized feedback, and tracks class performance using open-source NLP models and human-in-the-loop self-learning.

DocStream AI - We build an AI system that ingests unstructured documents such as PDFs, scans, and emails—understands the information by extracting key entities and relationships, and then automates entire business workflows end-to-end using large language models and graph-based reasoning.

Research Papers Implementations:

Decision Tree Stream – A Python implementation of a Research Paper, robust decision trees with node merging for better generalization. Supports both classification and regression.

Predicting the direction of stock market prices using random forest – A complete Python implementation of stock market movement prediction using Random Forest, inspired by a published research paper. This project covers preprocessing, technical indicator extraction, custom ensemble modeling, and visual evaluation for financial data.

Technical Skills

• **Programming & Development:**

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|------------------------|-----------------|-------------------|
| • Proficient in Python | • Scikit-Learn | • TensorFlow |
| • Keras | • PyTorch | • NLTK |
| • Transformers | • EDA | • ETL |
| • Matplotlib | • Plotly | • OpenCV |
| • Flask | • Django | • Streamlit |
| • Docker | • GitHub Action | • CI/CD pipelines |
| • MLflow | • DSA | • Problem Solving |

• **DataBases:** SQL, Vector Databases, MongoDB, PostgreSQL, neo4j

• **Data Science & AI:**

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|--------------------|------------------------|-----------------------|
| • Expertise in ML | • Statistical Modeling | • Data Visualization |
| • Deep Learning | • Data Analysis | • Predictive Modeling |
| • Data Engineering | • Cloud ML | • NLP |
| • Model Deployment | • Prompt Engineering | • Generative AI |
| • LangChain | • Llama-Index | • RAG |
| • LLM Finetuning | | |

• **Cloud Services:** AWS Cloud, Microsoft Azure, Google Cloud

• **Communication & Collaboration:** Strong in Effective Communication, Collaboration and Team Building.

• **Languages:** Fluent in English and Urdu with full professional proficiency.

Education

Bachelors In Information Technology

2018-2022

Python, DSA, OOP, Software Engineering